

CUSTOMER BEHAVIOR ANALYSIS

Customer Segmentation, Purchase Patterns & Churn Risk Analysis

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Dataset: Kaggle – Customer Behavior Analysis

1. Executive Summary

This project analyzes 250,000 customer transactions to understand customer value, purchasing patterns, retention and churn risk. RFM analysis was used to segment 49,673 unique customers into Champions, Loyal Customers, Potential Loyalists, At-Risk Customers and Lost / Hibernating customers. The results show a very high repeat-customer rate of 96.61%, but also a large inactive segment and a meaningful churn rate, creating clear opportunities for targeted engagement.

2. Dataset & Data Cleaning

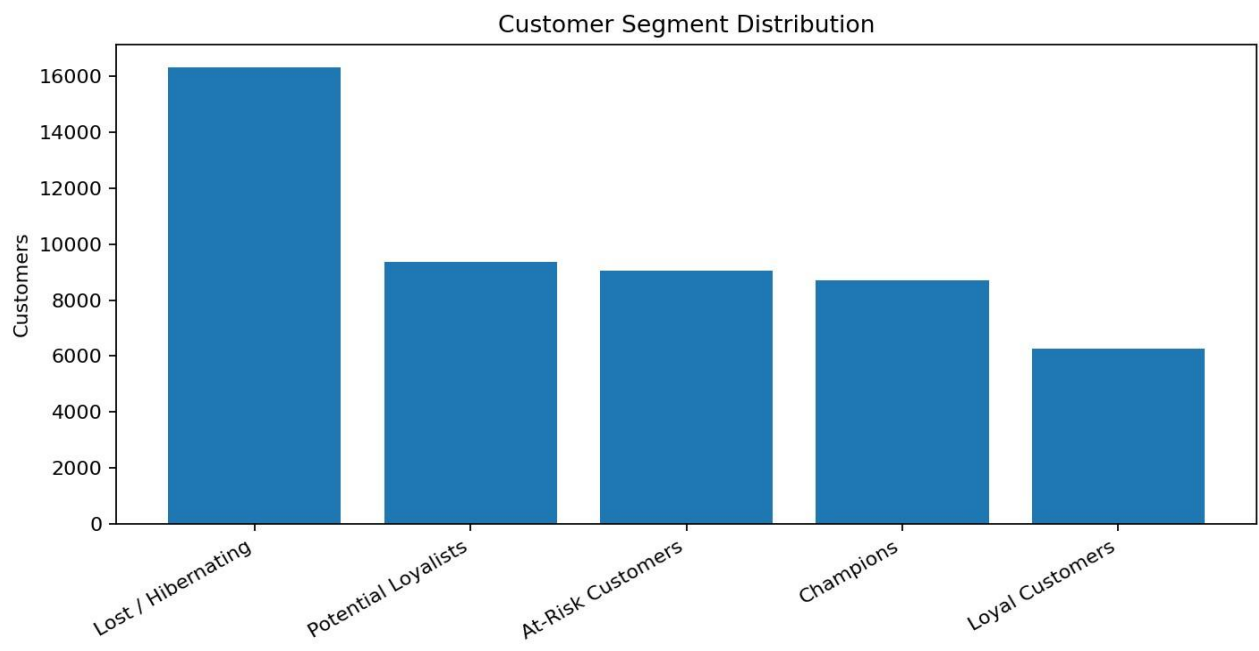
The dataset contains 250,000 transactions and 13 fields, including Customer ID, Purchase Date, Product Category, Product Price, Quantity, Total Purchase Amount, Payment Method, Returns, Age, Gender and Churn. There were no duplicate rows. Returns contained 47,596 missing values (19.04%); these were treated as no-return indicators for the analysis. Purchase Date was converted to datetime and essential analytical fields were checked before customer-level aggregation.

3. Feature Engineering & RFM

Recency measures days since the latest purchase, Frequency measures transaction count, and Monetary measures total purchase value. Customers were scored from 1 to 5 for each RFM dimension using quantiles, with higher scores representing more desirable behavior.

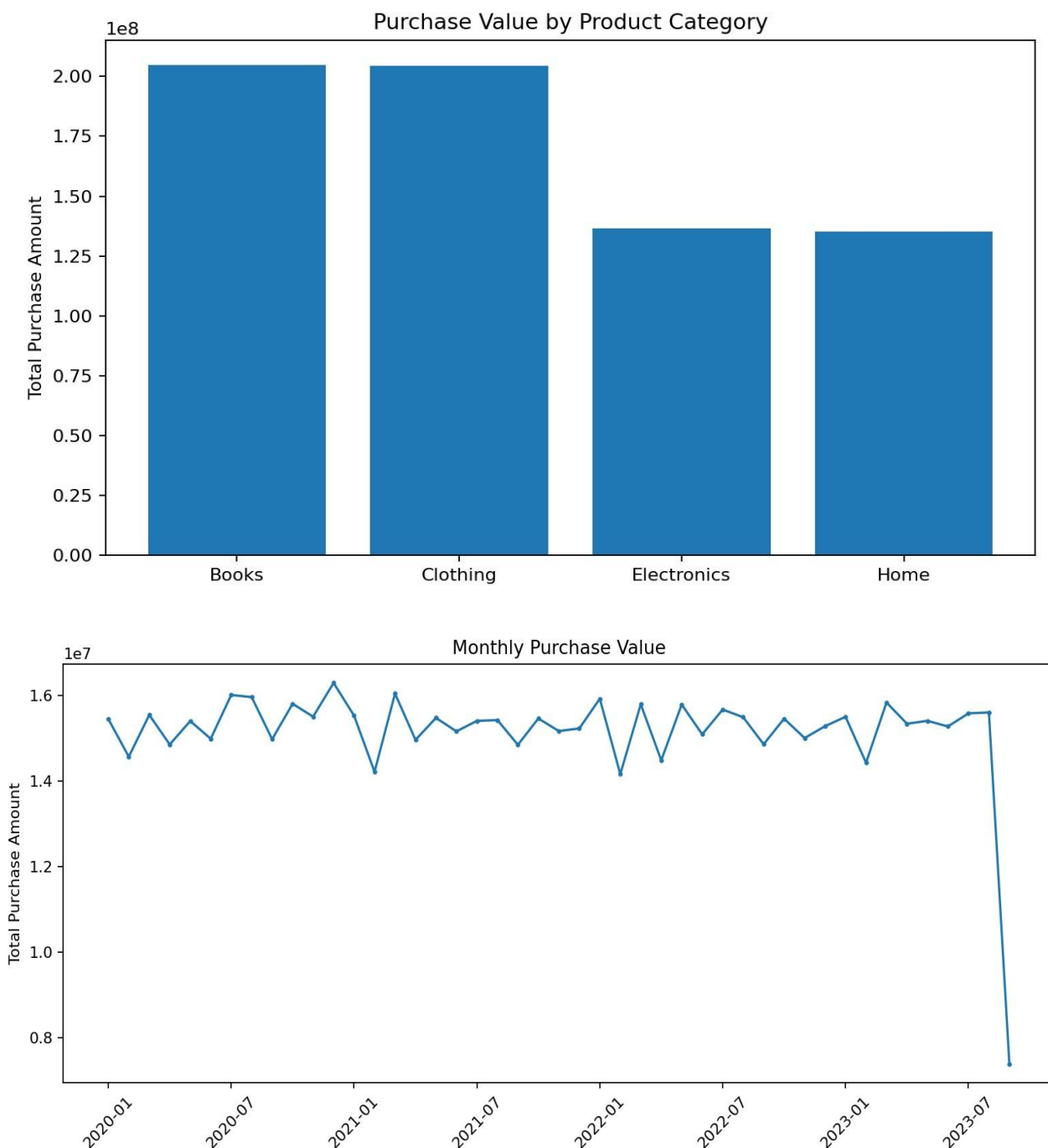
4. Customer Segments

Segment	Customers	Share	Avg. Monetary
Lost / Hibernating	16,316	32.85%	8,058.78
Potential Loyalists	9,348	18.82%	10,673.98
At-Risk Customers	9,049	18.22%	16,714.46
Champions	8,692	17.50%	21,789.39
Loyal Customers	6,268	12.62%	17,458.95



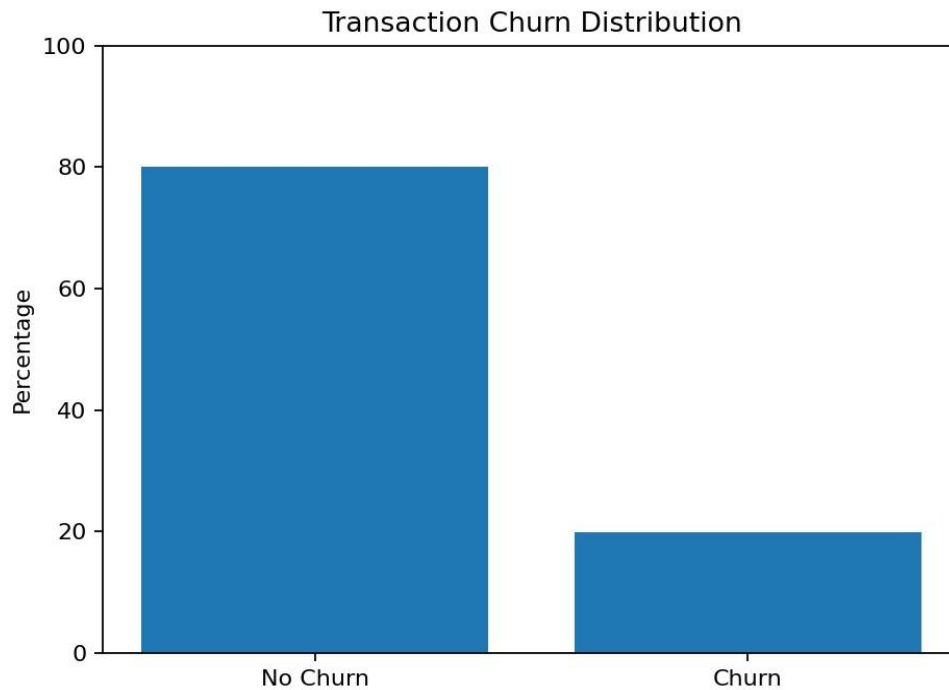
5. Purchase Patterns

Total purchase value across the observed period is approximately 681.34 million. Books generated the highest category purchase value (204.94 million), followed closely by Clothing (204.53 million). The highest monthly purchase value was December 2020 at approximately 16.29 million. September 2023 is lower because the source data ends part-way through that month.



6. Churn & Retention

The transaction-level churn proportion is 19.95%, while the customer-level churn rate, calculated as the maximum churn flag per customer, is approximately 20.01%. At the same time, 47,988 of 49,673 customers are repeat customers, giving a repeat-customer rate of 96.61%. High-value behavioral churn risk was also identified by combining high recency with above-median monetary value.



7. Key Findings

- Lost / Hibernating is the largest segment at 32.85% of customers.
- Champions are 17.50% of customers and have the highest average monetary value (21,789.39).
- At-Risk Customers are 18.22% of the base and have high average monetary value (16,714.46), making them available win-back target.
- Potential Loyalists represent 18.82% and are an opportunity for conversion into loyal customers.
- Repeat purchasing is very strong at 96.61%, but churn remains approximately 20.01%, so retention efforts should be targeted rather than broad.

8. Five Actionable Recommendations

Protect Champions: Use loyalty tiers, exclusive benefits, early access and personalized cross-sell offers.

Win back At-Risk Customers: Trigger personalized reminders, discounts and time-limited offers based on inactivity and previous value.

Convert Potential Loyalists: Encourage the next purchase with bundles, product recommendations and second-purchase incentives.

Reactivate Lost / Hibernating Customers: Run targeted low-cost win-back campaigns and reduce spend on persistently unresponsive customers.

Optimize category and campaign timing: Prioritize high-value categories such as Books and Clothing and use monthly purchase patterns to schedule campaigns.

9. Conclusion

The analysis provides Alfido Tech with a practical customer-engagement framework. RFM identifies who is most valuable, churn analysis identifies who needs attention, and purchase trends indicate where and when engagement can be improved. The recommended strategy is to protect high-value customers, convert promising customers and selectively reactivate inactive or at-risk customers.

Source: Kaggle Customer Behavior Analysis dataset supplied by the project author. Numerical findings in this report are based on the submitted notebook outputs.